Bachelor Thesis Presentation

University of Vienna
Faculty of Computer Science

Enhancing ProcessIntel

Infrequent Inductive Mining, XES Data Integration, and Static Graph Visualization

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Structure

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Goals of Thesis

01 Inductive Miner Infrequent

Integrating existing Inductive Miner Infrequent into the project structure to filter rare behavior

02 XES import and conversion

Integrating XES file support and CSV to XES and XES to CSV conversion, evaluating XES conversion against PM4Py

03 Static graph layout

Introducing new graph layout to preserve graph layout during filtering

04 Bug fixes

Fixing BPMN graph visualization behavior
Fixing column-name merging bug in XES to CSV conversion

Basics

Event Log

An event log contains events grouped into cases. Each event represents an activity, and timestamps are used to reconstruct the order of events within a case [1].

Process Discovery

Process discovery takes an event log containing example behaviors and automatically constructs a process model that describes the underlying process, such as a Directly-Follows Graph, Petri net, or BPMN model [1].

Visualization

Visualization supports the exploration and understanding of process models and event-log behavior. It can also present information such as bottlenecks, performance, data, and resources [2].

Basics: Inductive Miner Infrequent

Inductive Miner (IM) is a process discovery technique that creates a process tree. It recursively splits the initial log into smaller sublogs using four process operators and creates Directly-Follows Graph for each sublog [3].

Inductive Miner Infrequent (IMf) extends the behavior of IM by filtering infrequent behavior during process discovery. Filtering is applied directly to the Directly-Follows-Graph, log splitting, base cases and empty traces [3].

A filtering threshold controls how much of infrequent behavior is removed. Higher thresholds filter more rare behavior, while lower thresholds preserve more of the original log [3].

Case ID	Activity	Timestamp
1	Order Created	2026-01-10 09:00
1	Package Packed	2026-01-10 09:15
1	Shipped via Carrier	2026-01-11 13:00
1	Delivered	2026-01-15 13:00
2	Order Created	2026-01-12 10:00
2	Order Cancelled	2026-01-12 13:00

Figure 1: Event Log

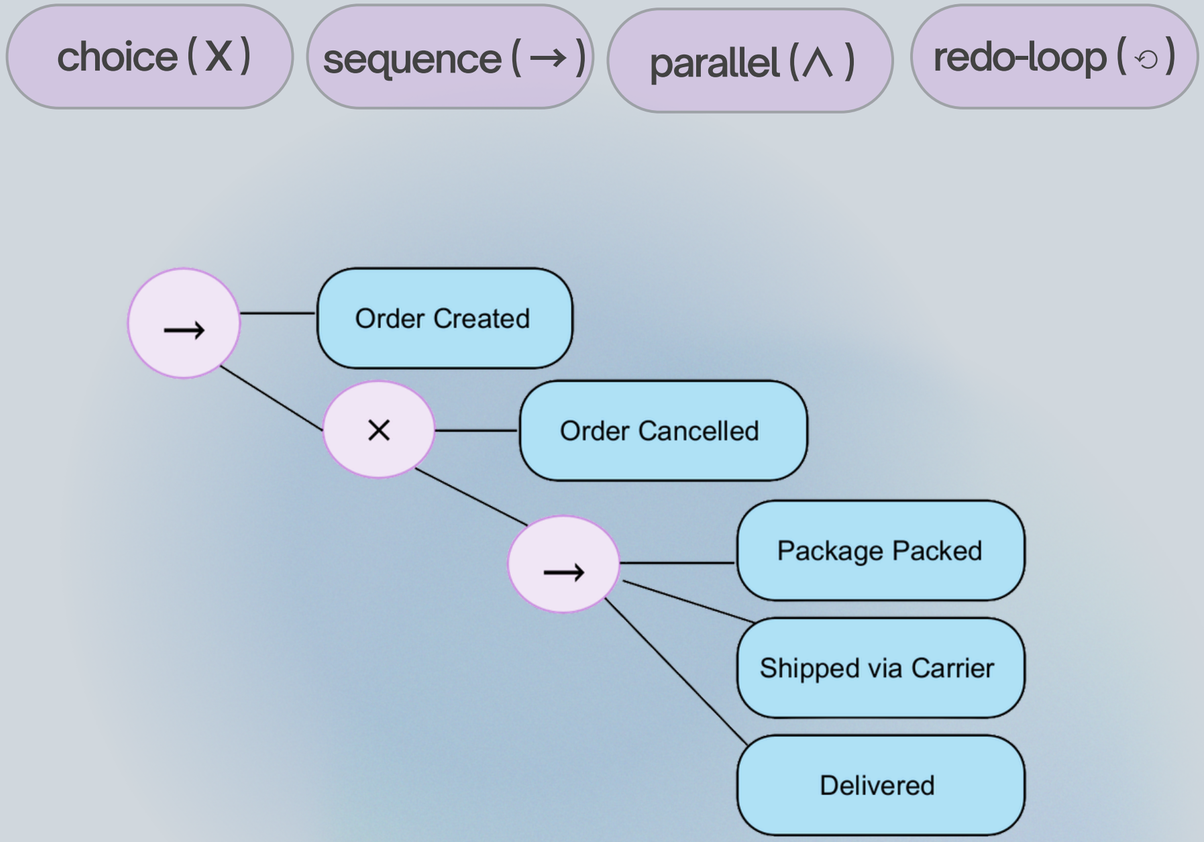
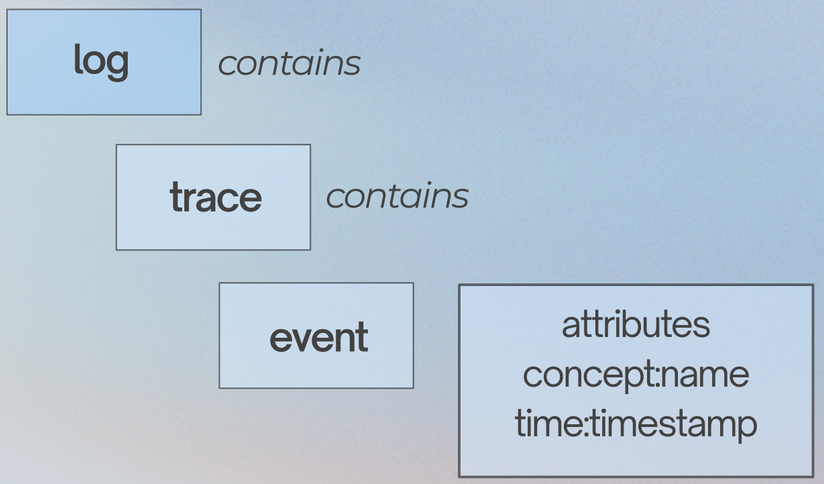


Figure 2: Constructed process tree

Basics: XES file format

- **XES (eXtensible Event Stream)** is an XML-based standard format for recording and sharing event logs between process mining tools [4] [5].
- XES uses a hierarchical structure: log → traces → events [4].
- Namespace (‘xmlns="http://www.xes-standard.org/"‘).
- XES supports typed attributes, including strings, dates, integers, floating-point values, collection types and other attribute types [5].




```xml
<log xmlns="http://www.xes-standard.org/"
     xes.version="1.0">

  <trace>
    <string key="concept:name" value="Case 1"/>

    <event>
      <string key="concept:name" value="Receive order"/>
      <date key="time:timestamp"
            value="2026-01-10T10:00:00+01:00"/>
    </event>

    <event>
      <string key="concept:name" value="Ship order"/>
      <date key="time:timestamp"
            value="2026-01-11T14:30:00+01:00"/>
    </event>
  </trace>

</log>
```

Figure 3: Example of XES file structure

Implementation: System Architecture

Six main components:

- View
- Controller
- Mining Model
- Graph Converter
- Import /Export
- Visualization

User → Streamlit [7] View → Controller → Mining Model → Graph Converter → Visualization → Streamlit View

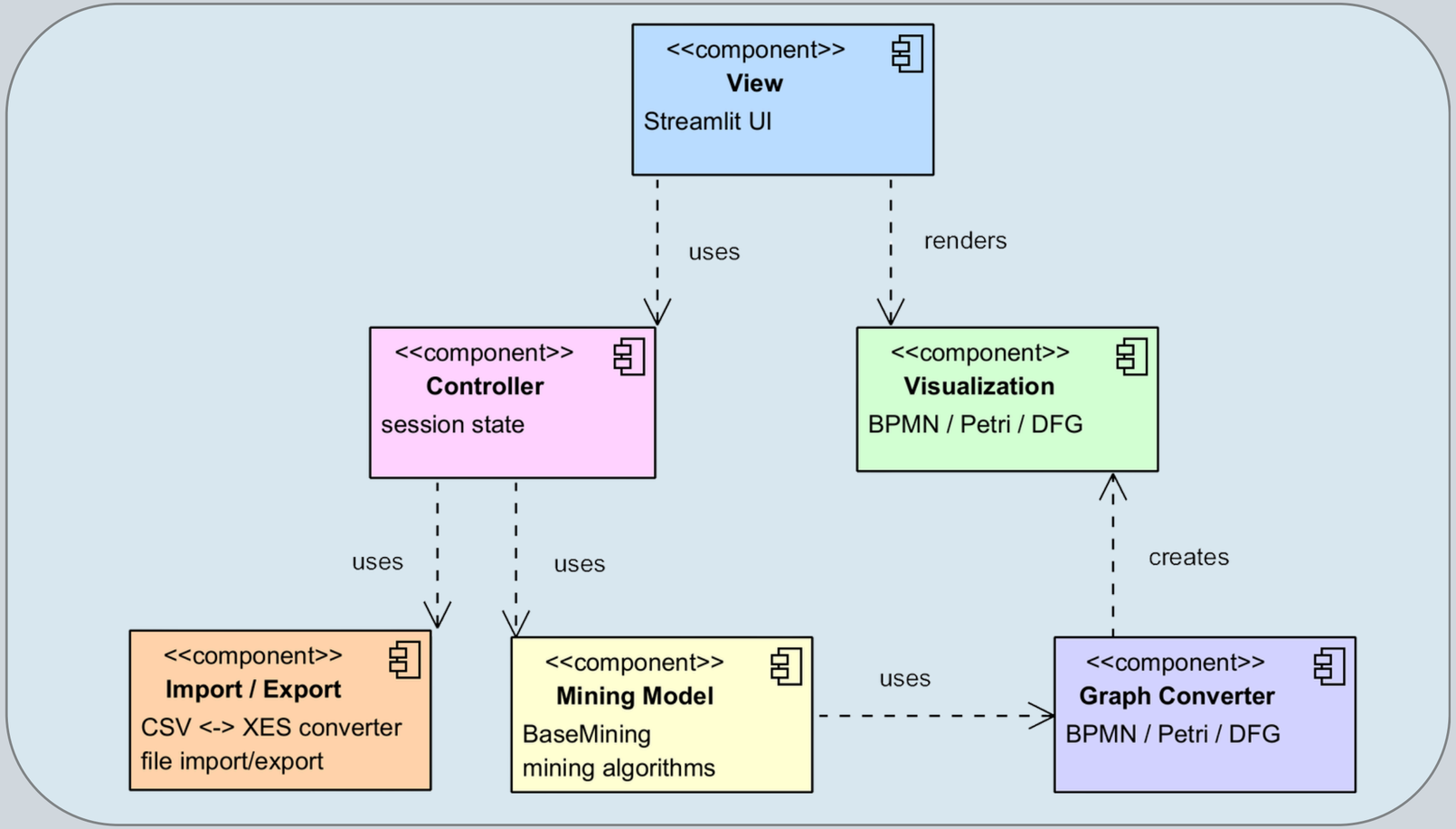
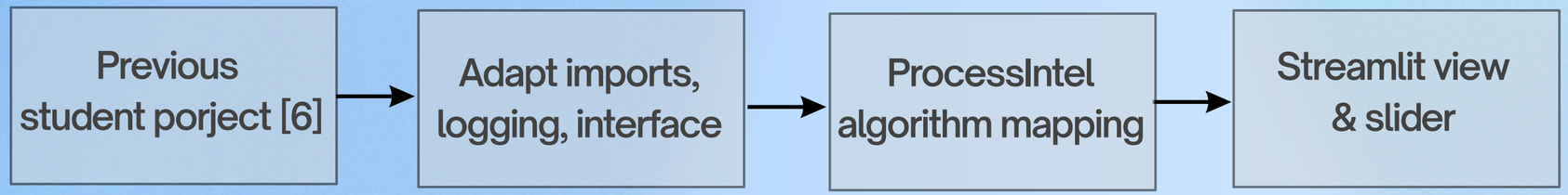


Figure 4: Component Diagram of System Architecture

Implementation: Integration of Inductive Miner Infrequent

Adapting an existing algorithm to the current arhcitecture:



Noise threshold controls how much infrequent behavior is removed during discovery.

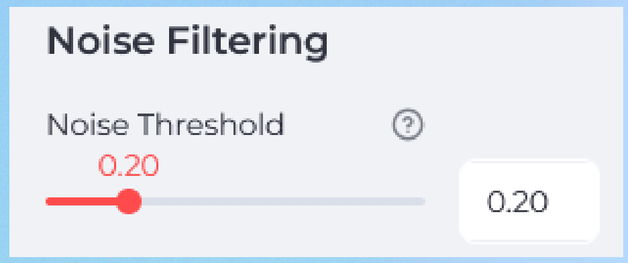


Figure 5: Noise threshold set to the default value

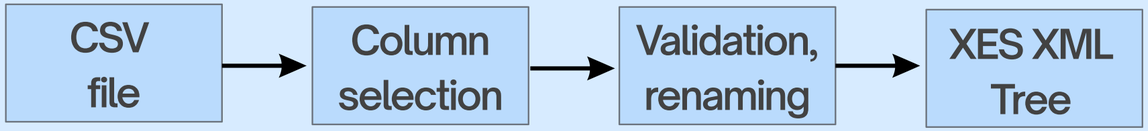
Bug Fix:
When filtering removed all activities, BPMN graph generation failed. The fix creates a valid graph.

```
    @staticmethod
    def _build_empty_graph(data: InductiveBPMNData) -> BPMNGraph:
        graph = BPMNGraph(data.filtered_events)
        graph.add_start_node()
        graph.add_end_node()
        graph.add_edge("Start", "End", None)
        return graph
```

Listing 2: Empty BPMN Graph creation method

Implementation: XES import and conversion

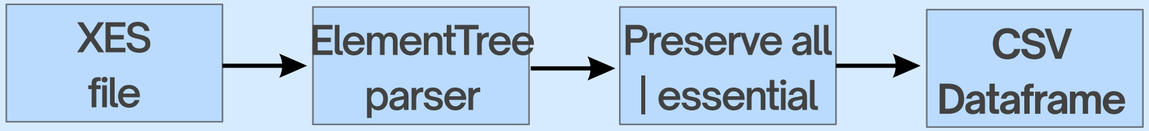
CSV → XES



- XES import support: '.xes' files were added to the import configuration and parsed into an 'ElementTree'
- Column handling: CSV case ID, activity, and timestamp columns are validated and mapped to XES attributes

- **Bug fix:** Existing 'concept:name' columns are preserved by renaming them with an 'original:' prefix instead of merging them with the selected activity column

XES → CSV

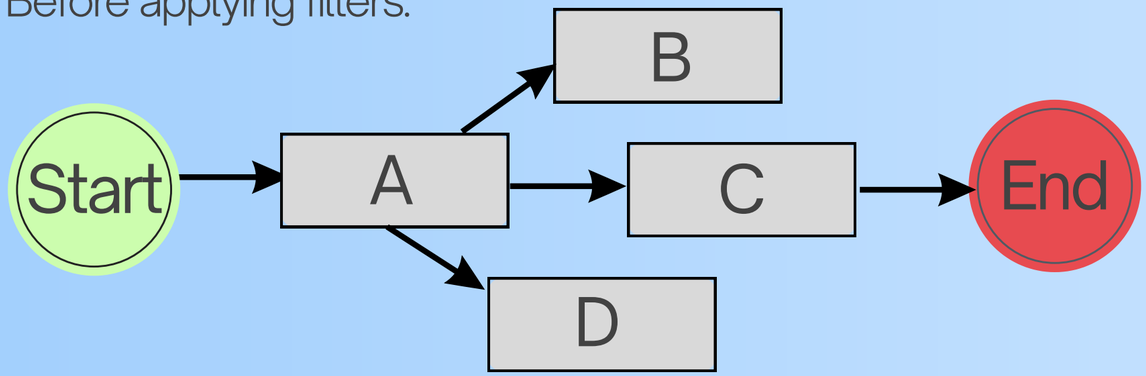



```python
def _get_original_name(self, column_name: str, existing_columns: list[
    str]) -> str:
    original_name = f"original:{column_name}"
    if original_name not in existing_columns:
        return original_name
    counter = 1
    while f"{original_name}:{counter}" in existing_columns:
        counter += 1
    return f"{original_name}:{counter}"
```

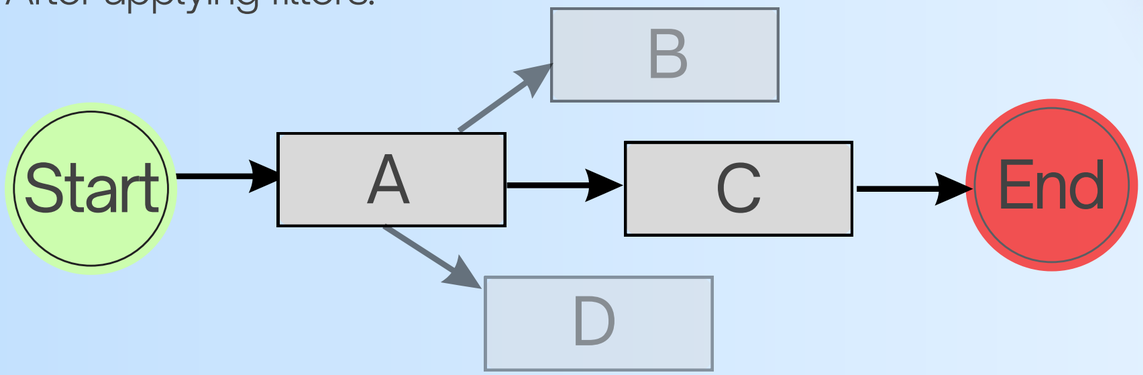
Listing 1: Creating Original Column Names

Implementation: static graph layout

Before applying filters:



After applying filters:



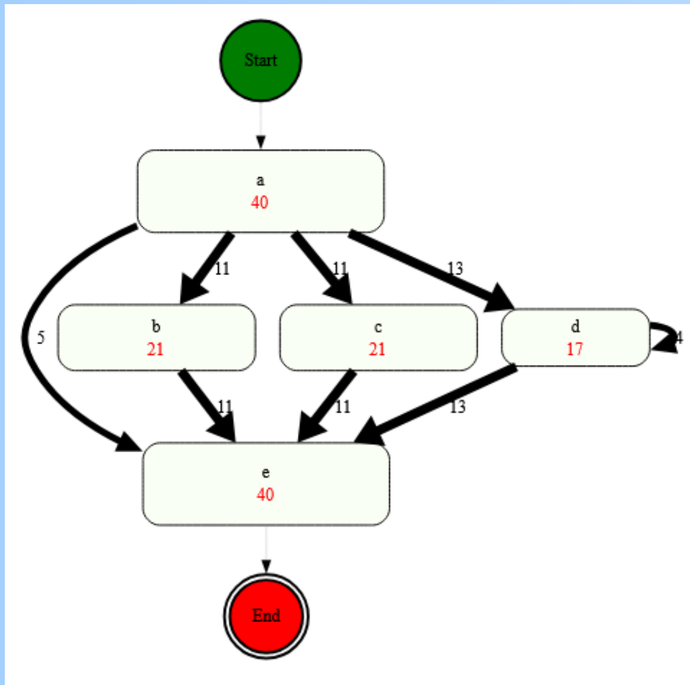


Figure 6: Static layout enabled, fillers not applied

Core logic:

```
fixed_graph = st.session_state.get("graph_fixed_position")

fixed_graph.set_invisible_nodes(graph.get_node_ids())
fixed_graph.update_visible_nodes(graph.get_nodes())
fixed_graph.set_invisible_edges(graph.get_edges())

return fixed_graph
```

Listing 3: Applying static graph layout

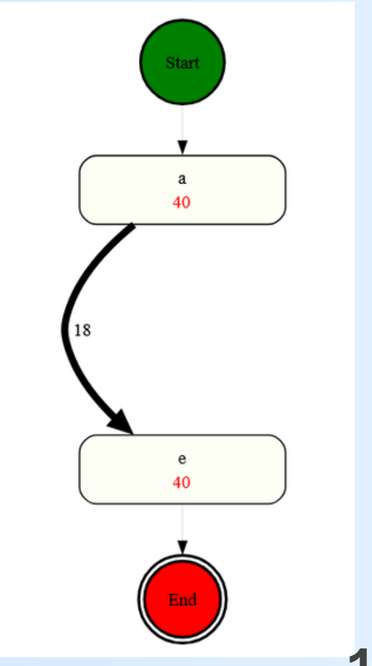


Figure 7: Static layout enabled, filters applied

Project Demonstration

01 Inductive Miner Infrequent

02 Conversion of CSV ↔XES

03 Static graph layout

Evaluation

- Manual UI testing
- Unit testing
- PR review

Area	Evaluation
XES conversion	PM4Py comparison: case IDs, activities, timestamps and event count matched for evaluated input.
‘concept:name’ bug	Test based on BPI Challenge 2013 behavior, existing attributes preserved.
IMf integration	Previous tests migrated and adapted to current architecture.
BPMN edge case	All activities filtered case now returns correct graph Start → End.
Static layout	BaseGraph unit tests + UI checks for invisible nodes and edges.

Limitations and Future Work

Limitations

Static layout supports only DFG-based algorithms

Edge weights are not recalculated while static layout is enabled

Large XES logs can still require noticeable processing time

PM4Py [9] evaluation covers selected CSV-to-XES cases only

Future Work

Extend static layout to BPMN graphs and Petri nets

Recalculate edge weights correctly while preserving positions

Validate XES-to-CSV conversion against additional datasets

Improve performance for large event logs

Open issues are listed in project repository: https://github.com/SWISDATA/processintel/issues

Summary and Lessons learned

Main contributions:

Integrated IMf into the current ProcessIntel architecture

Fixed conversion and BPMN visualization edge cases

Added XES support and CSV/XES conversion workflow

Added static layout support for filtering comparison

Lessons Learned:

- Working in an existing open-source project requires reading and adapting code before changing it.
- Small, reviewable pull requests and tests matter as much as feature itselt.
- Practical development involved Python [10], Streamlit [7], Graphviz [8], XML/XES, PM4Py [9] and Git workflows.
- A useful feature must be correct, maintainable, and understandable for future contributors.

Thank You

for your time
and attention

Sources:

[1] van der Aalst, 2022, “Foundations of Process Discovery”
[2] Kriglstein et al., 2016, “Visual Analytics in Process Mining”
[3] Sander J. J. Leemans, Dirk Fahland, and Wil M. P. van der Aalst. Discovering block-structured process models from event logs containing infrequent behaviour. Business Process Management Workshops, LNCS 171, pp. 66–78, Springer, 2014.
[4]Christian W. Günther and H. M. W. Verbeek. XES – standard definition. Technical Report 1409, BPMcenter.org, 2014.
[5] IEEE. IEEE Standard for eXtensible Event Stream (XES) for Achieving Interoperability in Event Logs and Event Streams. IEEE Std 1849-2023, 2023.
[6] Previous ProcessIntel student project implementing Inductive Miner Infrequent: https://github.com/zaker1998/ProcessMiningVisualization_SS25_Dussaliyev_/tree/master, Accessed: 2026-09-18
[7] Streamlit. Streamlit. https://streamlit.io/, Accessed: 2026-09-18
[8] Graphviz. Graphviz. https://graphviz.org/. Accessed: 2026-09-03.
[9] Process Intelligence Solutions GmbH. PM4Py – Process Mining for Python. https://processintelligence.solutions/pm4py, Accessed: 2026-09-18
[10] Python Software Foundation. Python Documentation. https://docs.python.org/3/, Accessed: 2026-09-18